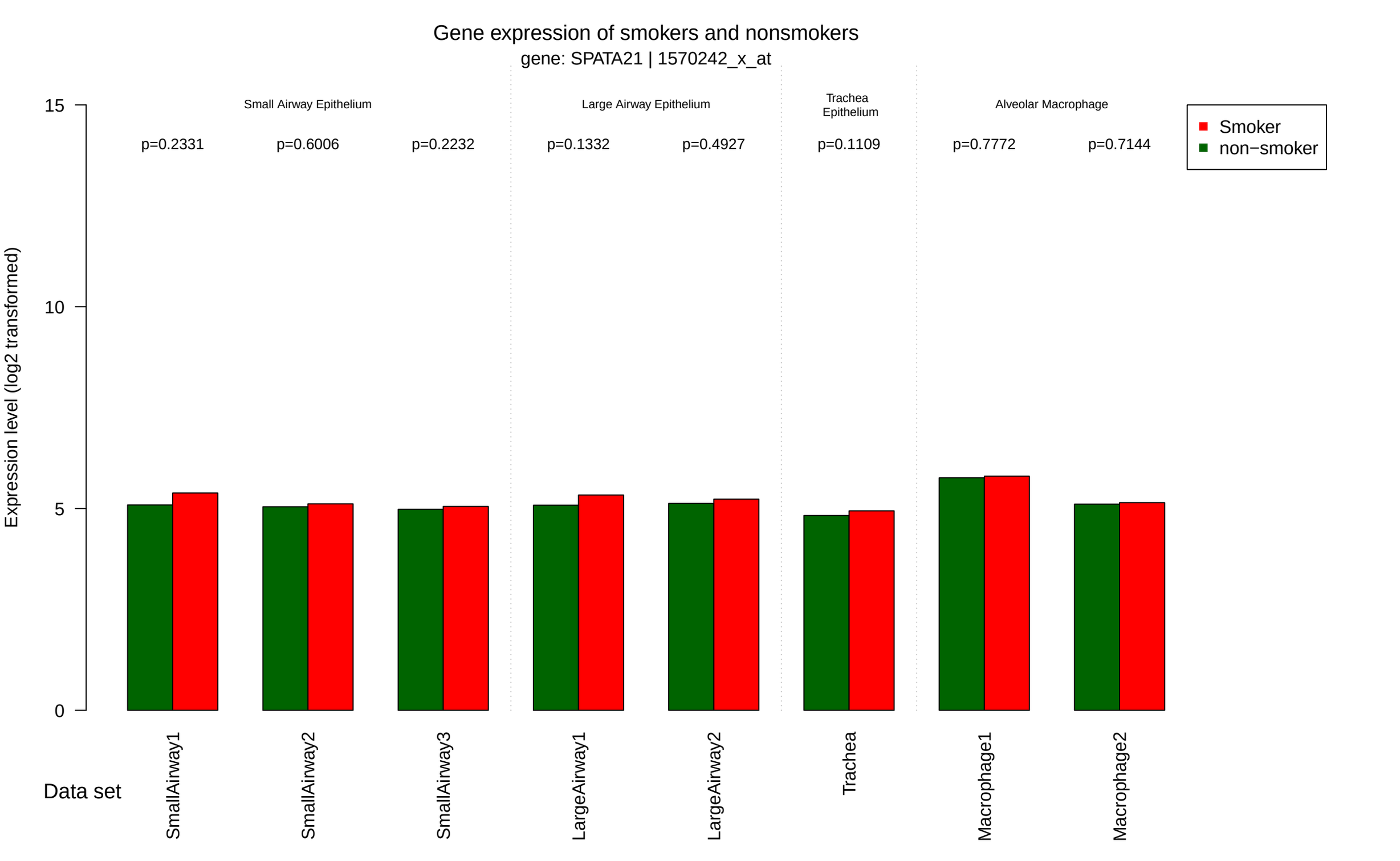
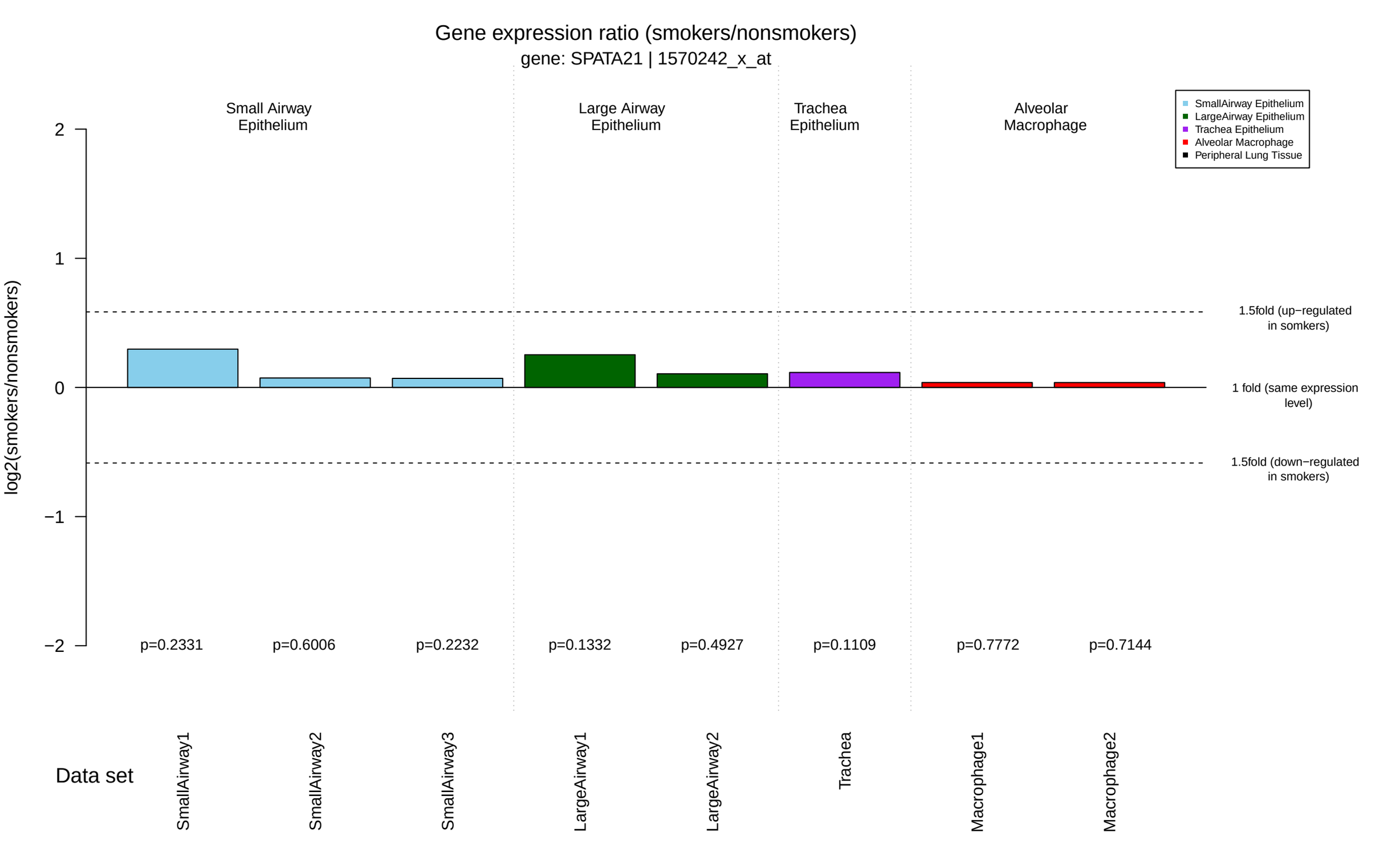
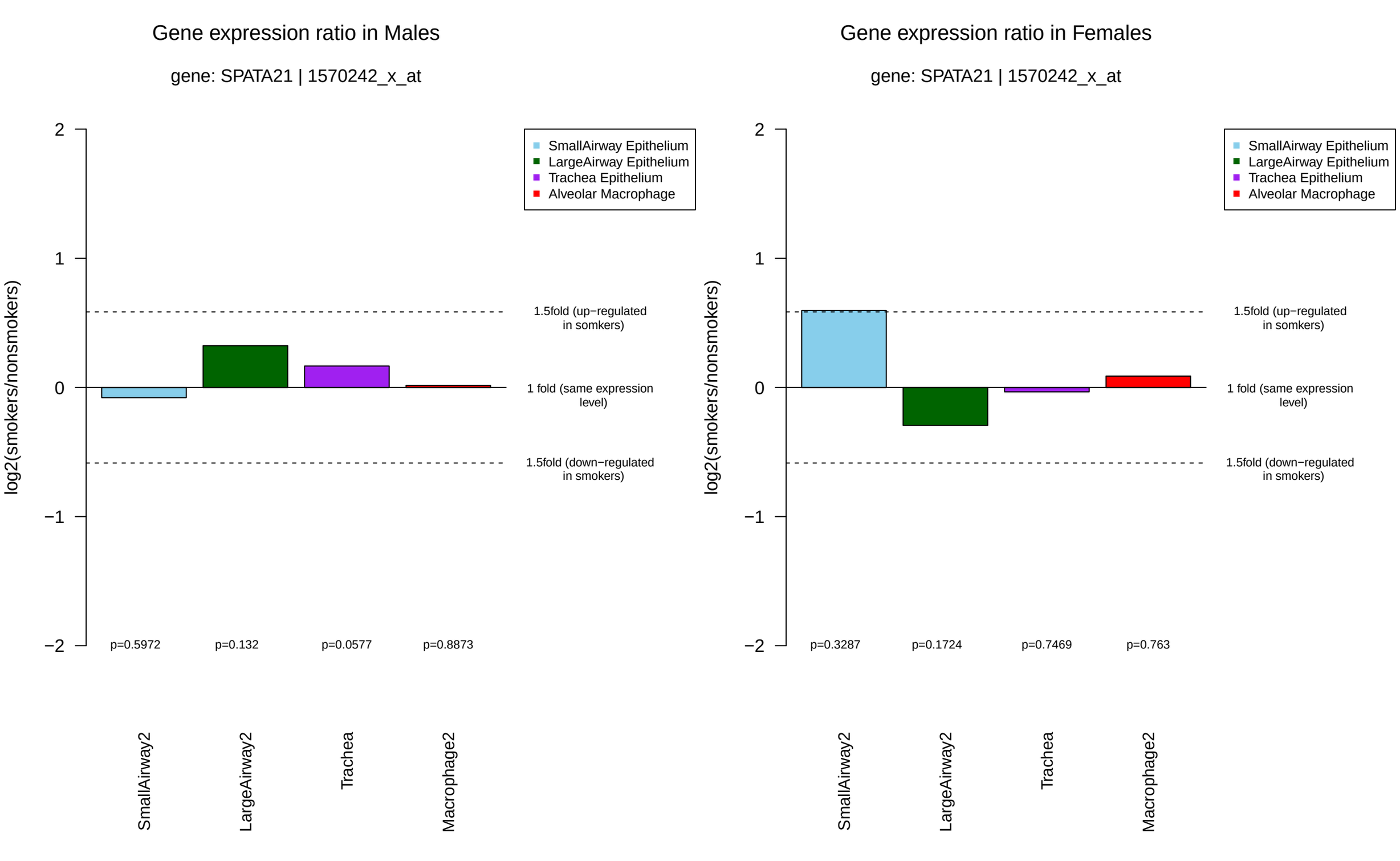
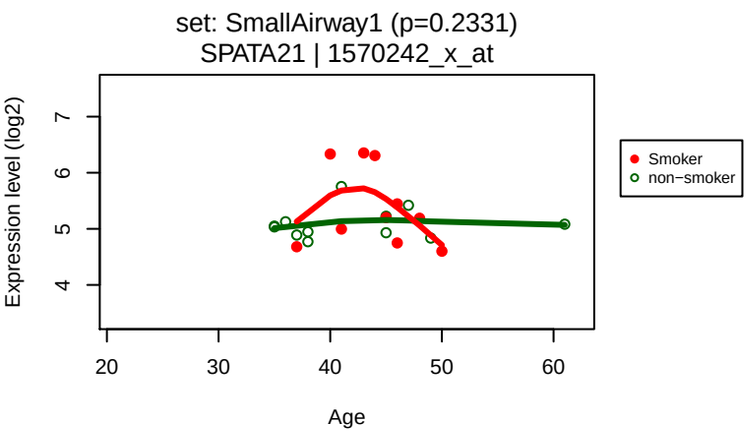
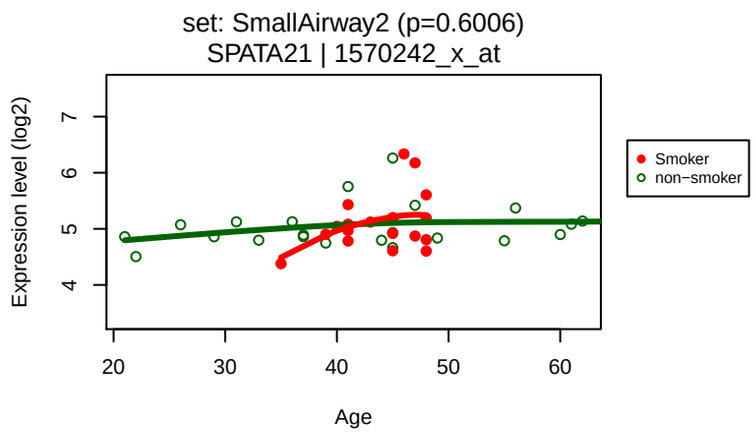
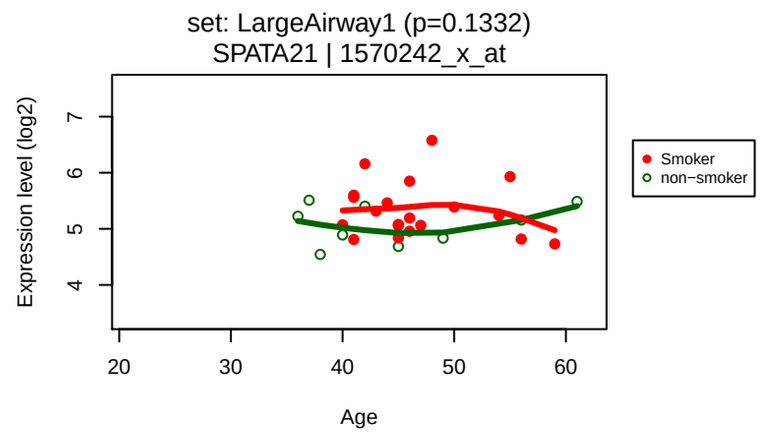

Gene expression of smokers and nonsmokers
gene: SPATA21 | 1570242_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2331
p=0.6006
p=0.2232
p=0.1332
p=0.4927
p=0.1109
p=0.7772
p=0.7144
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

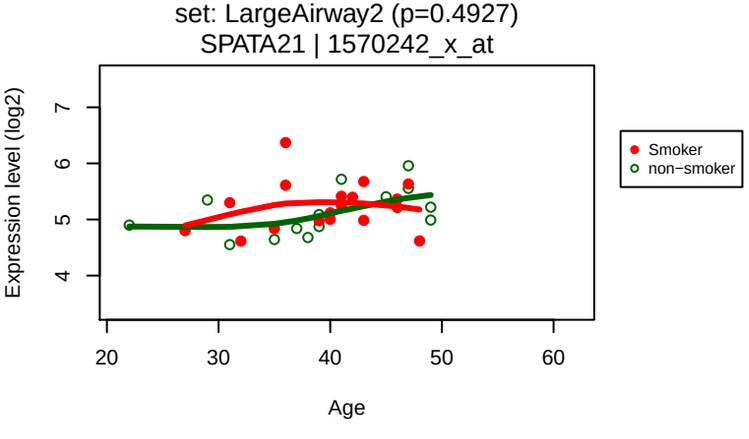

Gene expression ratio (smokers/nonsmokers)
gene: SPATA21 | 1570242_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2331
p=0.6006
p=0.2232
p=0.1332
p=0.4927
p=0.1109
p=0.7772
p=0.7144
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

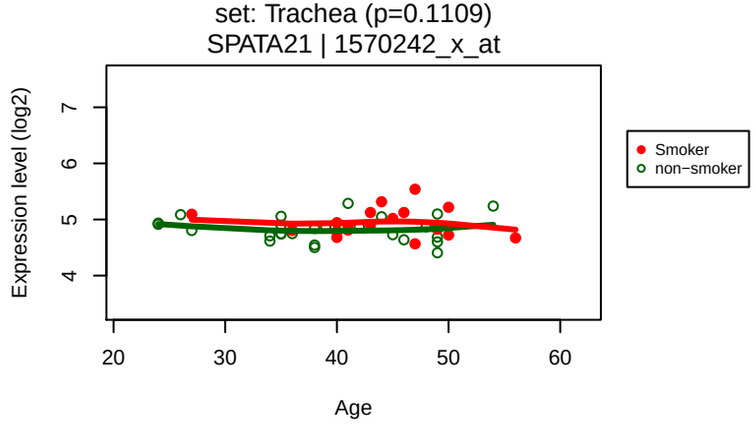

Gene expression ratio in Males
gene: SPATA21 | 1570242_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5972
p=0.132
p=0.0577
p=0.8873
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA21 | 1570242_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3287
p=0.1724
p=0.7469
p=0.763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

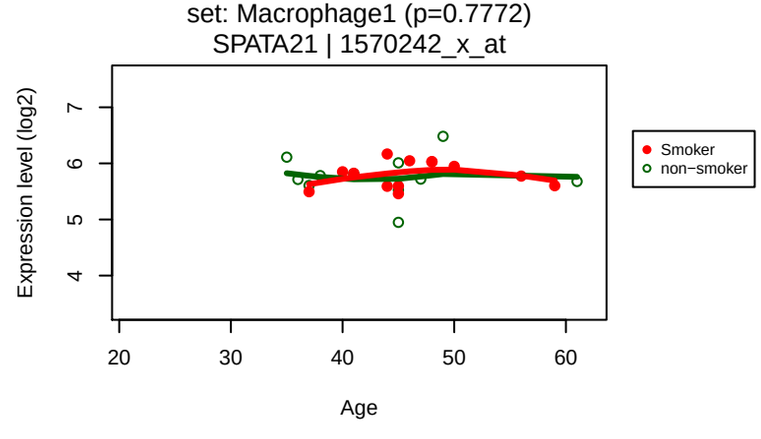

set: SmallAirway1 (p=0.2331)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker

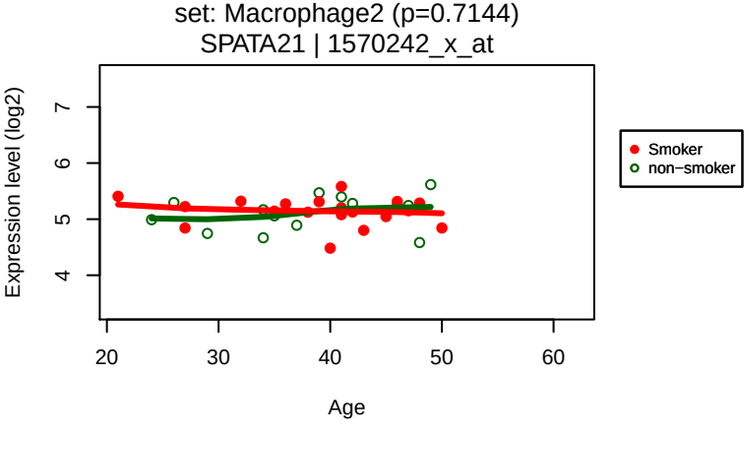

set: SmallAirway2 (p=0.6006)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker

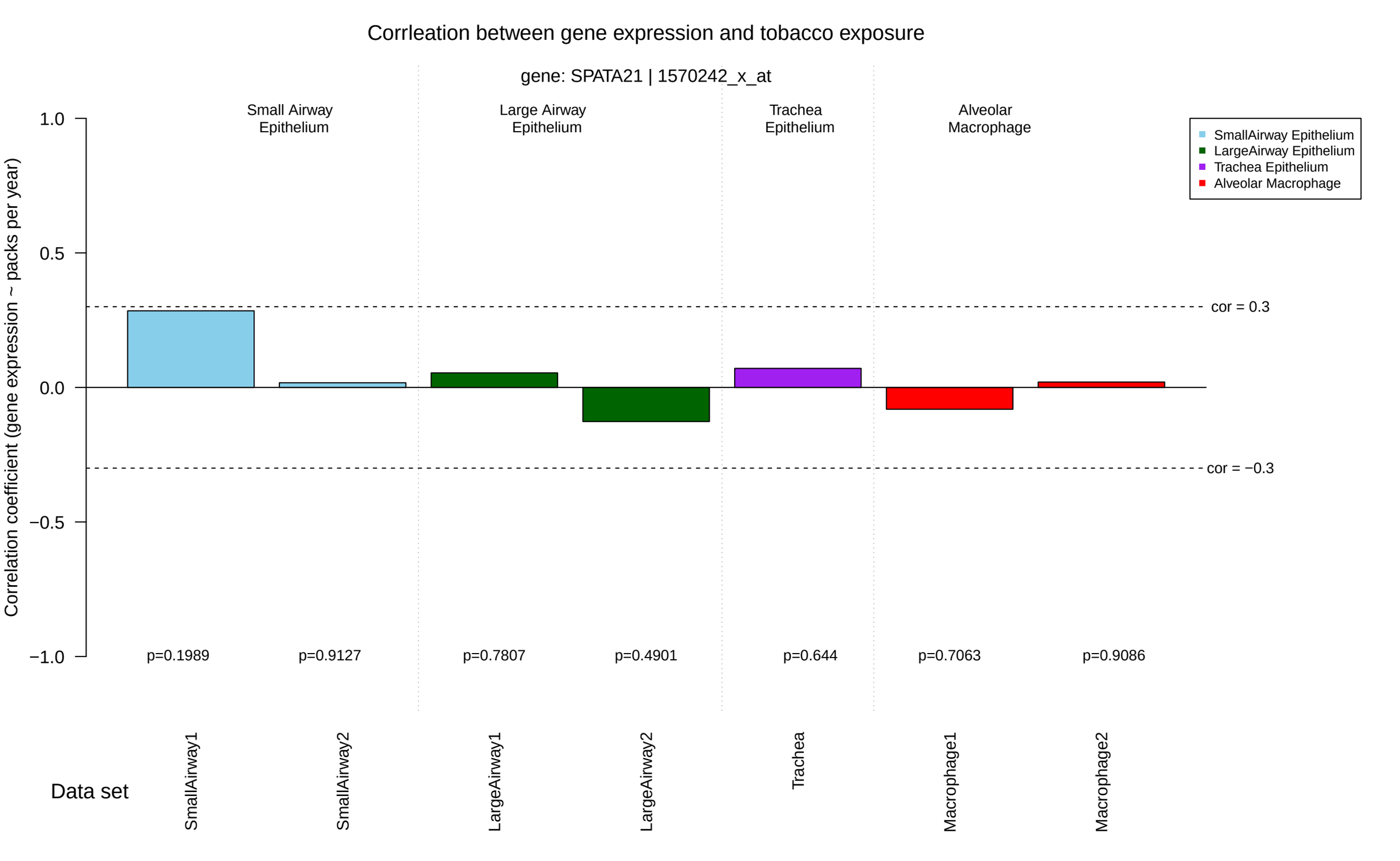

set: LargeAirway1 (p=0.1332)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4927)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1109)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7772)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7144)
SPATA21 | 1570242_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA21 | 1570242_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1989
p=0.9127
p=0.7807
p=0.4901
p=0.644
p=0.7063
p=0.9086
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2